# Function Notation

Exam code: 0607

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# Function Notation

## What is a function?

- A **function** is a combination of **one or more mathematical operations** that takes a **set of numbers** and **changes** them into **another set of numbers**
- The numbers being **put into the function** are often called the **inputs**
- The numbers **coming out of the function** are often called the **outputs**
- A function may be thought of as a mathematical "**machine**"
  - For example, for the function "double the number and add 1", the two mathematical operations are "multiply by 2 (×2)" and "add 1 (+1)"
    - Putting 3 in to the function would give 2 × 3 + 1 = 7
    - Putting -4 in would give 2 × (-4) + 1 = -7
    - Putting $x$ in would give $2x + 1$

## What is function notation?

- A function, f, with input $x$ can be written as $f(x) = ...$
  - Letters other than f can be used
    - The letters g, h and j are common but any letter can be used
    - Typically, a **new letter** will be used to define a **new function** in a question
- For example, the function with the rule "triple the number and subtract 4" would be written
  - $f(x) = 3x - 4$
- In such cases, "$x$" is the **input** and "$f(x)$" is the **output**
- Sometimes functions don't have names like f and are just written as $y = ...$
  - E.g. $y = 3x - 4$

## How does a function work?

- A function has an **input** $x$ and an **output** $f(x)$
  - If the input is 2, then the output is $f(2)$
  - If the input is $m$, then the output is $f(m)$
  - If the input is $t + 5$, then the output is $f(t + 5)$
    - You cannot simplify this output any further
- If the **function is known**, the **output can be calculated**

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- $f(3) = 2 \times 3 + 1 = 7$
- $f(-4) = 2 \times (-4) + 1 = -7$
- $f(a) = 2a + 1$

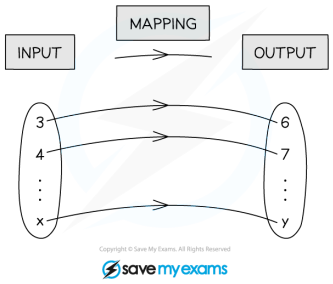
- If the **output is known**, an **equation** can be formed and solved to **find the input**
  - For example, given the function $f(x) = 2x + 1$
    - If $f(x) = 15$, then form an equation by replacing $f(x)$ with $2x + 1$
    - $2x + 1 = 15$
    - Solving this equation gives an input of 7

- Note that $f(x) = 15$ and $f(15)$ are very different things:
  - $f(x) = 15$ means an input of $x$ gives an output of 15
  - $f(15)$ means substitute the input 15 into the function

## What is a mapping diagram?

- A **mapping diagram** shows a **set of different inputs** going into the function to become a **set of different outputs**
  - Transforming inputs into outputs is called **mapping**

- For example, a mapping diagram for the function $f(x) = x + 3$ where $x \geq 3$ could be shown as:




### Worked Example

A function is defined as $f(x) = 3x^2 - 2x + 1$.

Find $f(7)$.

The input is $x = 7$, so substitute 7 into the expression everywhere you see an $x$

$$f(7) = 3(7)^2 - 2(7) + 1$$

Calculate

$$\begin{aligned} f(7) &= 3(49) - 14 + 1 \\ &= 147 - 14 + 1 \end{aligned}$$

$$\mathbf{f(7) = 134}$$

### Worked Example

This is not saying substitute 16 into the function
It says that an input $x$ is substituted into g giving the output -16

To find the input, form an equation by replacing $g(x)$ with $3x - 4$

$$3x - 4 = -16$$

Solve the equation (for example, by adding 4 to both sides, then dividing by 3)

$$3x - 4 = -16$$
$$3x = -12$$
$$x = -\frac{12}{3}$$

$$x = -4$$

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